

Power BI Assignment - 1

Data Transformation & Data Modeling

Data Import & Transformation:

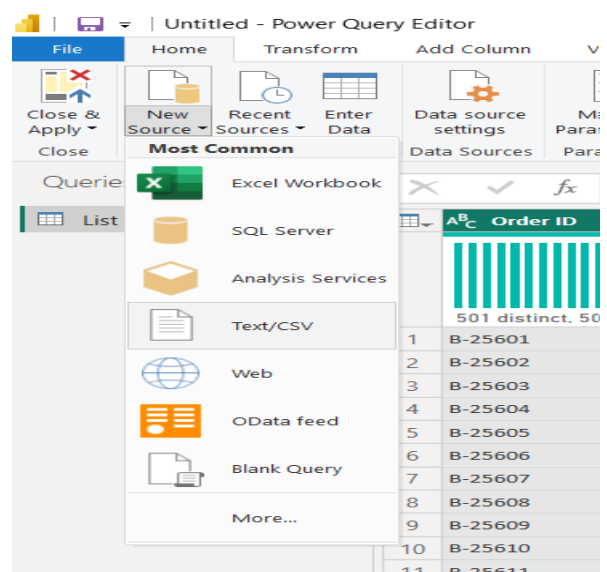
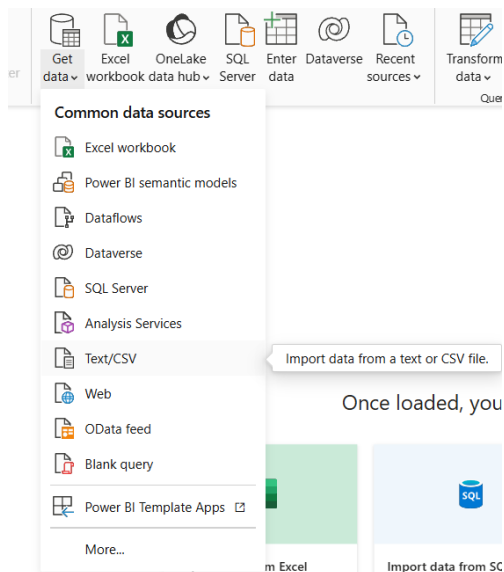
1. Data Import:

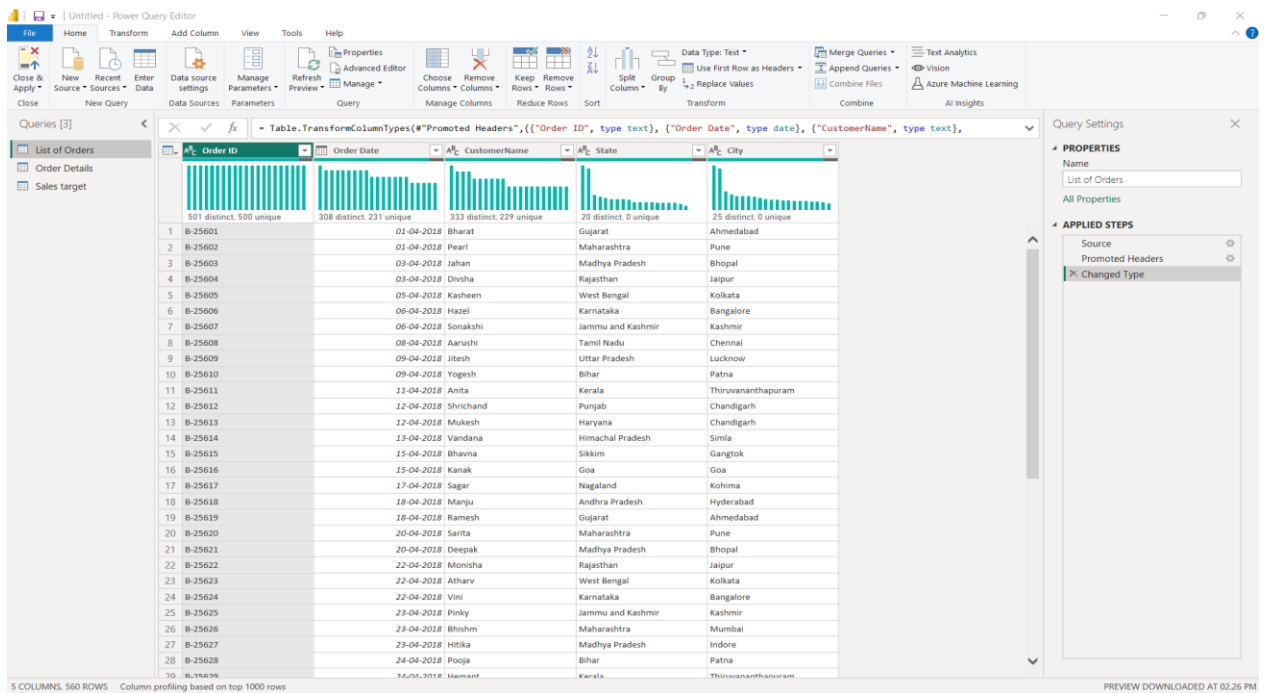
- Import List of Orders.csv into Power BI.
- Open the list of orders in the Power Query Editor using the Transform Data option.
- Import Order Details.csv and Sales Target.csv into Power Query Editor.

ANS:

The datasets List of Orders, Order Details, and Sales Target were imported into Power BI using the Get Data option

All datasets were loaded into Power Query Editor for further transformation and preparation.



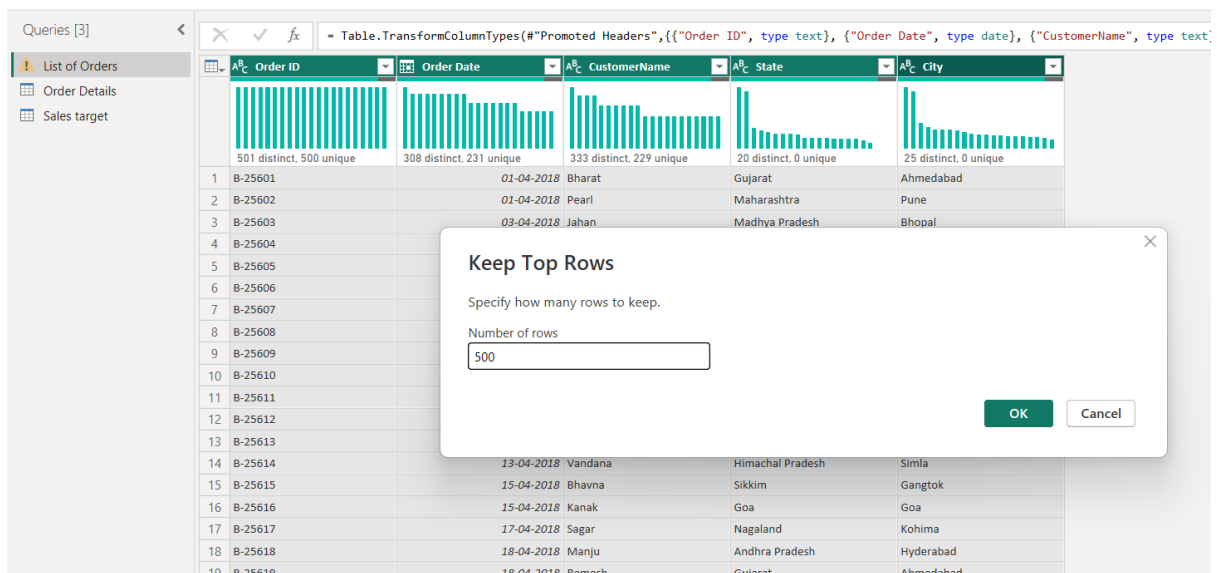


2. Row Limiting & Data Types:

- o Restrict the List of Orders table to the first 500 rows.

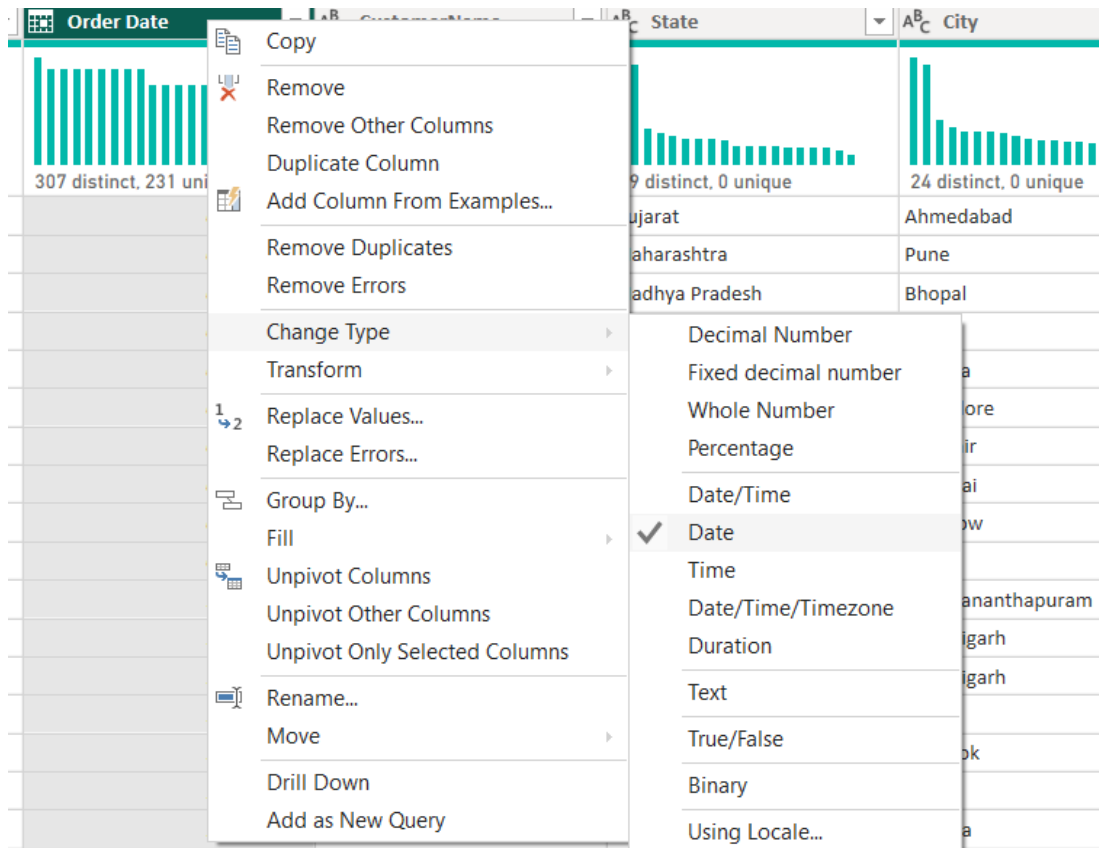
ANS:

The List of Orders table was restricted to the first 500 rows using the Keep Top Rows option in Power Query Editor



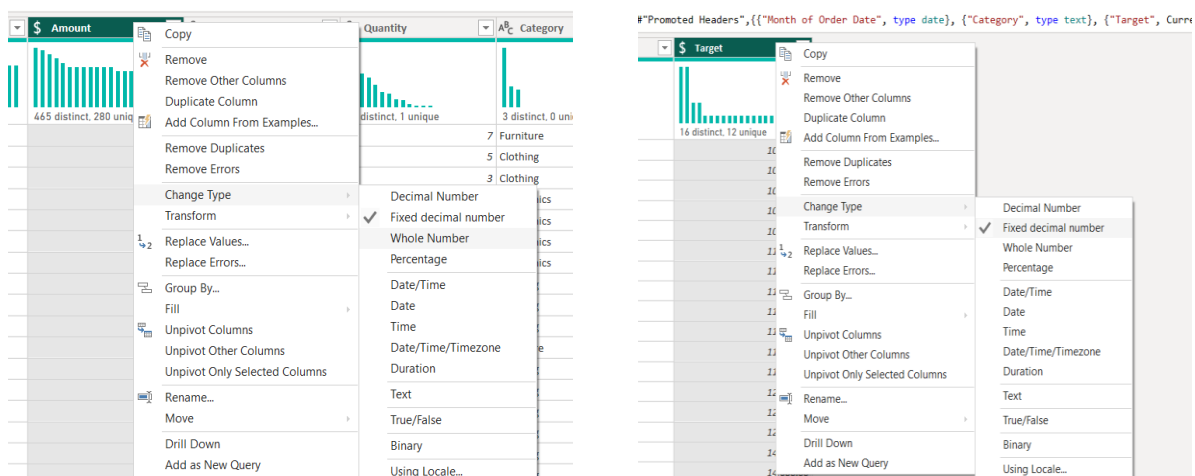
o Set the Order Date column to the Date data type.

ANS: The Order Date column was converted into Date format by using change type option



o Change the Amount and Target columns to fixed decimal numbers.

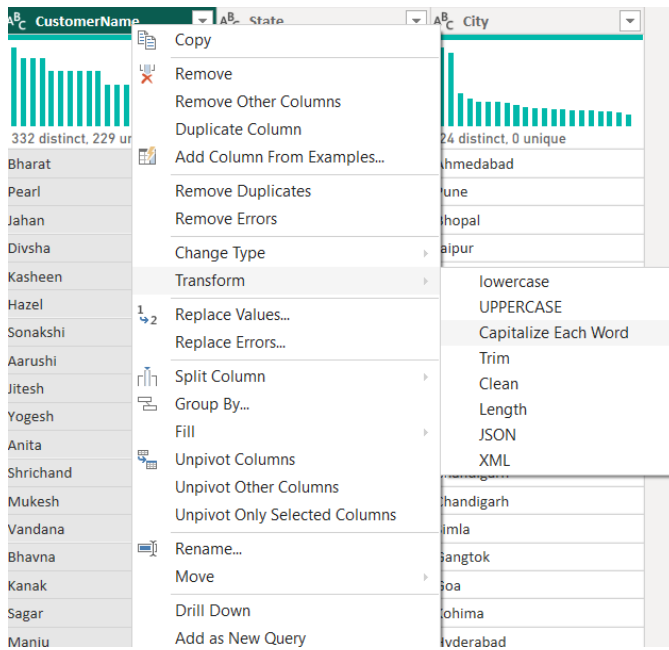
ANS: The Amount and Target columns were changed to Fixed Decimal Number format



3. Text Formatting:

o Format the Customer Name column to proper case to ensure consistent capitalization.

ANS: The CustomerName column was formatted using the Capitalize Each Word option to ensure consistent text formatting and proper capitalization throughout the dataset

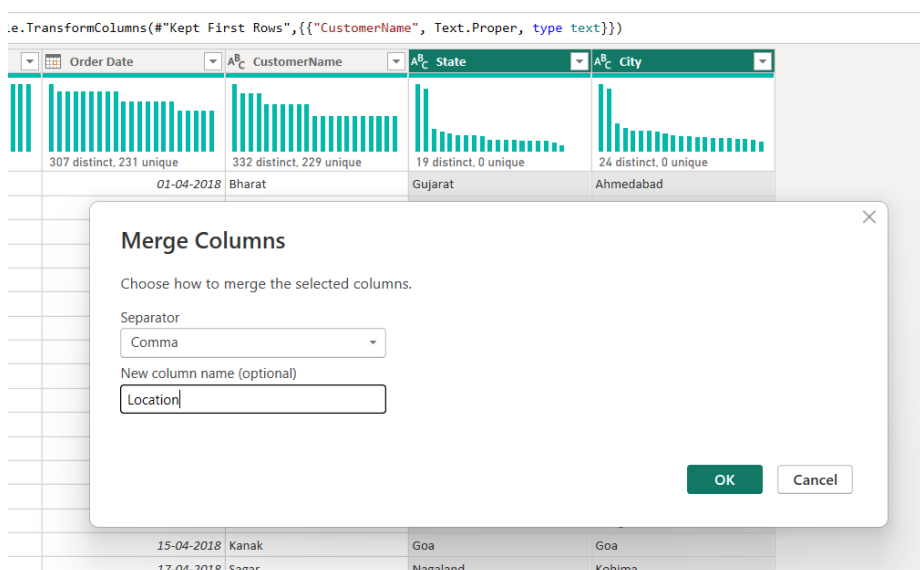


4. Column Creation:

o Merge the City and State columns to create a new column named Location in the format:

City, State

ANS: The City and State columns were merged to create a new column named Location in the format "City, State"



Queries [3] Table.CombineColumns(#"Capitalized Each Word",{"City", "State"},Combiner.CombineTextByDelimiter(", ", QuoteStyle.None),"Location")

	Order ID	Order Date	CustomerName	Location
1	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat
2	B-25602	01-04-2018	Pearl	Pune,Maharashtra
3	B-25603	03-04-2018	Jahan	Bhopal,Madhya Pradesh
4	B-25604	03-04-2018	Divsha	Jaipur,Rajasthan
5	B-25605	05-04-2018	Kasheen	Kolkata,West Bengal
6	B-25606	06-04-2018	Hazel	Bangalore,Karnataka
7	B-25607	06-04-2018	Sonakshi	Kashmir,Jammu and Kashmir
8	B-25608	08-04-2018	Aarushi	Chennai,Tamil Nadu
9	B-25609	09-04-2018	Jitesh	Lucknow,Uttar Pradesh
10	B-25610	09-04-2018	Yogesh	Patna,Bihar
11	B-25611	11-04-2018	Anita	Thiruvananthapuram,Kerala
12	B-25612	12-04-2018	Shrichand	Chandigarh,Punjab
13	B-25613	12-04-2018	Mukesh	Chandigarh,Haryana
14	B-25614	13-04-2018	Vandana	Simla,Himachal Pradesh
15	B-25615	15-04-2018	Bhavna	Gangtok,Sikkim
16	B-25616	15-04-2018	Kanak	Goa,Goa
17	B-25617	17-04-2018	Sagar	Kohima,Nagaland
18	B-25618	18-04-2018	Manju	Hyderabad,Andhra Pradesh

Query Settings

PROPERTIES

Name
List of Orders

All Properties

APPLIED STEPS

Source
Promoted Headers
Changed Type
Kept First Rows
Capitalized Each Word
Merged Columns

o Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)

ANS: The Profit Margin column was created using a custom formula by dividing Profit by Amount the resulting values were converted into percentage format

General From text From number From date & time AI insights

Queries [3] Table.TransformColumnTypes(#"Promoted Headers",{"Order ID", type text}, {"Amount", Currency.Type}, {"Profit", Int64.Type})

	Order ID	Amount	Profit	Quantity	Category	Sub-Category
1	B-25601					
2	B-25601					
3	B-25601					
4	B-25601					
5	B-25602					
6	B-25602					
7	B-25602					
8	B-25602					
9	B-25602					
10	B-25603					
11	B-25603					
12	B-25603					
13	B-25603					
14	B-25603					
15	B-25603					
16	B-25603					
17	B-25603					
18	B-25604					
19	B-25604					

Custom Column

Add a column that is computed from the other columns.

New column name
Profit Margin

Custom column formula
=

Available columns
Order ID
Amount
Profit
Quantity
Category
Sub-Category

<< Insert

Learn about Power Query formulas

The formula is incomplete.

OK Cancel

Custom Column

Add a column that is computed from the other columns.

New column name
Profit Margin

Custom column formula
= [Profit]/[Amount]

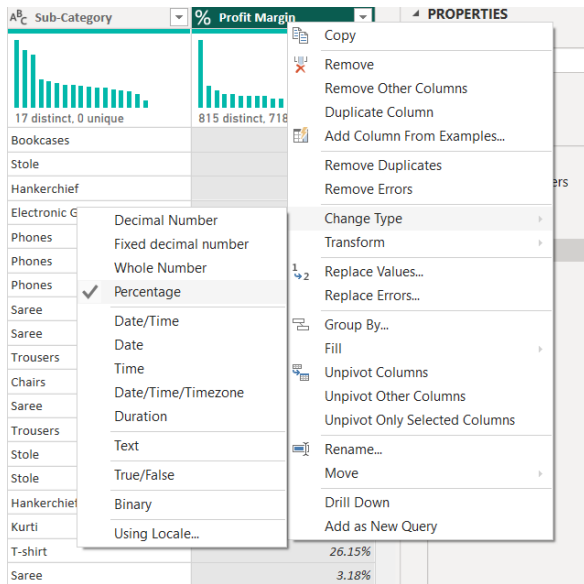
Available columns
Order ID
Amount
Profit
Quantity
Category
Sub-Category

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel



5. Conditional Column:

○ Add a conditional column named Profit Status based on:

■ Profit < 0 → Loss

■ Profit = 0 → Break-Even

■ Profit > 0 → Profit

ANS: A conditional column named Profit Status was created to categorize transactions as Profit, Loss, or Break-Even based on the profit values

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Profit Status

	Column Name	Operator	Value	Output
If	Profit	is less than	0	loss
Else If	Profit	equals	0	Break-Even
Else If	Profit	is greater than	0	Profit

Add Clause

Else

OK Cancel

o Name the merged output table as Orders Data.

ANS: The merged output was renamed as Orders Data to combine order-level and product-level information into a single table for analysis

Table: RenameColumns(#"Expanded Order Details",{"Order Details.Amount", "Amount"}, {"Order Details.Profit", "Profit"}, {"Order Details.Quantity", "Quantity"}, {"Order Details.Category", "Category"}, {"Order Details.Sub-Category", "Sub-Category"})

	Location	Amount	Profit	Quantity	Category	Sub-Category
1	Ahmedabad,Gujarat	1,275.00	-1148	7	Furniture	Bookcases
2	Ahmedabad,Gujarat	66.00	-12	5	Clothing	Stole
3	Ahmedabad,Gujarat	8.00	-2	3	Clothing	Hankierchief
4	Ahmedabad,Gujarat	80.00	-56	4	Electronics	Electronic Games
5	Pune,Maharashtra	168.00	-111	2	Electronics	Phones
6	Pune,Maharashtra	424.00	-272	5	Electronics	Phones
7	Pune,Maharashtra	2,617.00	1151	4	Electronics	Phones
8	Pune,Maharashtra	561.00	212	3	Clothing	Saree
9	Pune,Maharashtra	119.00	-5	8	Clothing	Saree
10	Bhopal,Madhya Pradesh	1,355.00	-60	5	Clothing	Trousers
11	Bhopal,Madhya Pradesh	24.00	-30	1	Furniture	Chairs
12	Bhopal,Madhya Pradesh	193.00	-166	3	Clothing	Saree
13	Bhopal,Madhya Pradesh	180.00	5	3	Clothing	Trousers
14	Bhopal,Madhya Pradesh	116.00	16	4	Clothing	Stole
15	Bhopal,Madhya Pradesh	107.00	36	6	Clothing	Stole
16	Bhopal,Madhya Pradesh	12.00	1	2	Clothing	Hankierchief
17	Bhopal,Madhya Pradesh	38.00	18	1	Clothing	Kurti
18	Jaipur,Rajasthan	65.00	17	2	Clothing	T-shirt
19	Jaipur,Rajasthan	157.00	5	9	Clothing	Saree
20	Kolkata,West Bengal	75.00	0	7	Clothing	Saree
21	Bangalore,Karnataka	87.00	4	2	Clothing	Shirt
22	Kashmir,Jammu and Kashmir	50.00	15	4	Clothing	Leggings
23	Chennai,Tamil Nadu	1,364.00	-1864	5	Furniture	Tables
24	Chennai,Tamil Nadu	476.00	0	3	Furniture	Chairs
25	Chennai,Tamil Nadu	257.00	23	5	Clothing	Hankierchief
26	Chennai,Tamil Nadu	856.00	385	6	Electronics	Printers
27	Lucknow,Uttar Pradesh	485.00	29	4	Electronics	Electronic Games
28	Lucknow,Uttar Pradesh	25.00	-5	4	Clothing	Saree

7. Handling Missing Data & Duplicate Data:

- Identify missing values across all tables.

ANS:

All tables were inspected for missing values using column filters in Power Query Editor no null or missing values were identified in the dataset therefore no replacement or removal operations were required

Column Filter: Location

Filter: null

Remove Empty: ☒

Text Filters: (Select All Search Results)

- Define and apply appropriate strategies such as:

- Replacing missing values

- Removing records

- Retaining nulls with justification

ANS:

Possible strategies such as replacing missing values, removing records, or retaining null values based on business relevance were considered during the validation process

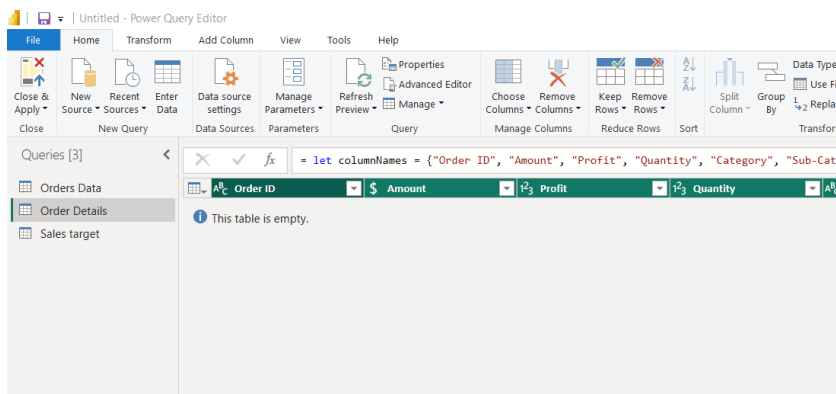
- Identify duplicate rows and define a suitable handling strategy based on business logic

ANS:

Duplicate rows were checked across all tables using the Keep Duplicates option in Power Query Editor no duplicate records were identified in the dataset; therefore, no duplicate removal operations were required

Order ID	Order Date	CustomerName	Location	Amount	Profit	Quantity
1	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	1,275.00	-1148
2	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	66.00	-12
3	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	8.00	-2
4	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	80.00	-56
5	B-25602	01-04-2018	Pearl	Pune,Maharashtra	168.00	-111
6	B-25602	01-04-2018	Pearl	Pune,Maharashtra	424.00	-272
7	B-25602	01-04-2018	Pearl	Pune,Maharashtra	2,617.00	1151
8	B-25602	01-04-2018	Pearl	Pune,Maharashtra	561.00	212
9	B-25602	01-04-2018	Pearl	Pune,Maharashtra	119.00	-5
10	B-25603	03-04-2018	Jahan	Bhopal,Madhya Pradesh	1,355.00	-60
11	B-25603	03-04-2018	Jahan	Bhopal,Madhya Pradesh	24.00	-30
12	B-25603	03-04-2018	Jahan	Bhopal,Madhya Pradesh	193.00	-166
13	B-25603	03-04-2018	Jahan	Bhopal,Madhya Pradesh	180.00	5
14	B-25603	03-04-2018	Jahan	Bhopal,Madhya Pradesh	116.00	16
15	B-25603	03-04-2018	Jahan	Bhopal,Madhya Pradesh	107.00	36
16	B-25603	03-04-2018	Jahan	Bhopal,Madhya Pradesh	12.00	1
17	B-25603	03-04-2018	Jahan	Bhopal,Madhya Pradesh	38.00	18
18	B-25604	03-04-2018	Divsha	Jaipur,Rajasthan	65.00	17

Order ID	Order Date	CustomerName	Location	Amount	Profit	Quantity	Category	Sub-Category
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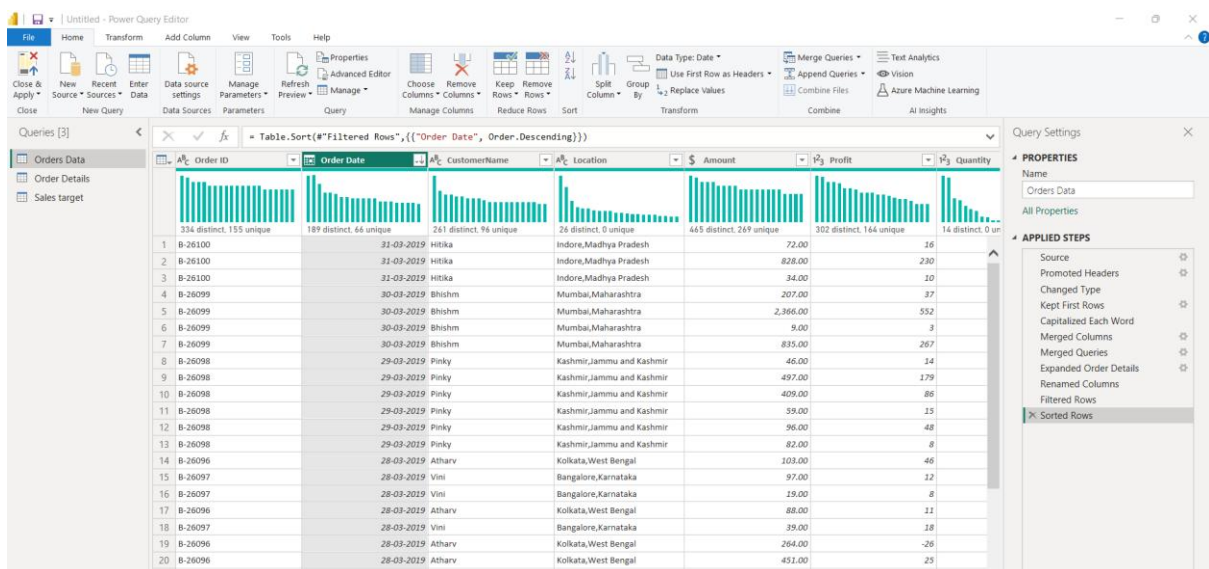


8. Sorting and Filtering Data:

- Sort records in the Orders Data table by Order Date in descending order.

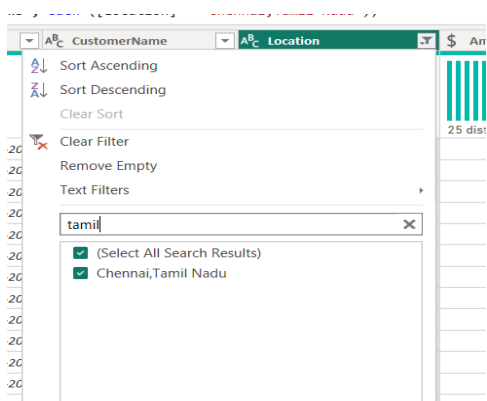
ANS:

Records in the Orders Data table were sorted by Order Date in descending order using the sorting options in Power Query Editor



- Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

ANS: Filters were applied on the Location column to analyze orders specifically from Tamil Nadu



9. Grouping and Aggregating Data:

- Duplicate the Order Details table and perform the following:
 - Count of Order ID

ANS:

The Order Details table was duplicated and grouped by Category to calculate the count of Order IDs using the Group By transformation in Power Query Editor

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Category

New column name

Order Count

Operation

Count Rows

Column

OK

Cancel

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Transpose Reverse Rows Detect Data Type Rename

Data Type: Whole Number Replace Values Fill Pivot Column Convert to List

Unpivot Columns Move Split Column Format Parse

Merge

Queries [4]

Orders Data Order Details Sales target Order Details Summary

Table Group(#"Filtered Rows", {"Category"}, {"Order Count"

	Category	Order Count
	3 distinct, 3 unique	3 distinct, 3 unique
1	Furniture	243
2	Clothing	949
3	Electronics	308

o Average Profit by Category

ANS:

The duplicated Order Details table was grouped by Category to calculate the average profit for each product category

×

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Category

New column name

Average Profit

Operation

Average

Column

Profit

OK

Cancel

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Queries [5]

- Orders Data
- Order Details
- Sales target
- Order Details Summary
- Average Profit Summary

Table

Category	Average Profit
3 distinct, 3 unique	3 distinct, 3 unique
1 Furniture	-497.4
2 Clothing	-5.764705882
3 Electronics	123

Formula Bar: = Table.Group(#"Filtered Rows1", {"Category"}, {"Average Profit", Average, null})

o Total Amount by Sub-Category

ANS: The Order Details table was grouped by Sub-Category to calculate the total sales amount for each sub-category using aggregation functions

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Sub-Category

New column name

Total Amount

Operation

Sum

Column

Amount

OK

Cancel

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Queries [6]

- Orders Data
- Order Details
- Sales target
- Order Details Summary
- Average Profit Summary
- SubCategory Amount Su...

fx = Table.Group(#"Filtered Rows", {"Sub-Cat

	Sub-Category	Total Amount
1	Bookcases	56861
2	Stole	18546
3	Hankerchief	14608
4	Electronic Games	39168
5	Phones	46119
6	Saree	53511
7	Trousers	30039
8	Chairs	34222
9	Kurti	3361
10	T-shirt	7382
11	Shirt	7555
12	Leggings	2106
13	Tables	22614
14	Printers	58252
15	Accessories	21728
16	Furnishings	13484
17	Skirt	1946

- Duplicate the Sales Target table and aggregate the total Target by Month of order date.

ANS: The Sales Target table was duplicated and grouped by Month of Order Date to calculate the total target values for each month

Group By

Specify the column to group by and the desired output.

☒ Basic
 ☐ Advanced

Month of Order Date

New column name

Operation

Column

Total Target

Sum

Target

OK

Cancel

Queries [7]

Orders Data

Order Details

Sales target

Order Details Summary

Average Profit Summary

SubCategory Amount Su...

Sales target Summary

Month of Order Date

1.2 Total Target

12 distinct, 12 unique

12 distinct, 12 unique

1	01-04-2018	31400
2	01-05-2018	31500
3	01-06-2018	31600
4	01-07-2018	33800
5	01-08-2018	33900
6	01-09-2018	34000
7	01-10-2018	36100
8	01-11-2018	36300
9	01-12-2018	36400
10	01-01-2019	43500
11	01-02-2019	43600
12	01-03-2019	43800

10. Data Modeling:

- Establish a relationship between:
 - List of Orders and Order Details using Order ID
- Establish a relationship between:
 - Order Details and Sales Target using Category

ANS: Relationships were established between the Orders Data and Order Details tables using the Order ID column.

Another relationship was created between the Order Details and Sales Target tables using the Category column.

Manage relationships ×					
+ New relationship		 Autodetect	 Edit  Delete  Filter ▼		
☐	From: table (column) ↑	Relationship	To: table (column)	Status	
☐	Order Details (Order ID)		Orders Data (Order ID)	Active	...
☐	Sales target (Category)		Order Details (Category)	Active	...

Use Manage Relationships to ensure relationships are active and correctly defined.

ANS: Many-to-many relationships were used because repeated Order ID and Category values were present across related tables as part of the transactional structure of the dataset

